Additional file 1 - Description of the multilevel repeated measures used in this study

$$Y_{tij} = \beta_0 + \beta_t x_t + \beta_i x_{0ij} + \beta_j z_j + v_j + U_{ti}$$

t = time 2006 .. 2008

i = individual 1 ..1048

j = neighbourhood 1 .. 259

Y_{tij} outcome score at moment (t) for an individual (i) in neighbourhood (j)

β_0 intercept

$\beta_t x_t$ deviation (β_t) from intercept for time (x_t)

$\beta_i x_{0ij}$ effect for variables measured at the individual level that are constant over time

$\beta_j z_j$ effect for neighbourhood variables that are constant over time

v_j random effect, the between neighbourhood variance

U_{ti} random effect, complete time dependent between and within individual

variance/covariance matrix $\begin{pmatrix} \sigma_{2006} & \sigma_{2006,2007} & \sigma_{2006,2008} \\ \sigma_{2006,2007} & \sigma_{2007} & \sigma_{2007,2008} \\ \sigma_{2006,2008} & \sigma_{2007,2008} & \sigma_{2008} \end{pmatrix}$

on the diagonal we have the within individual yearly (error) variance
and off the diagonal the between individual covariance between two years.
All parameters of this matrix are estimated from the data and are unstructured, meaning they can all be different. All the random effects are assumed to be normally distributed.
Note that this variance/covariance matrix is modelled at the level of the individuals, the lowest level, the individual measurements, is empty.